

Culturally-Historically Responsive Literacy Mini-Unit:
Exponential Functions in Our World

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Unit Context

This mini unit was designed for a fictional 9th grade Regents algebra 1 class. While I am not currently teaching this course, I am a long-term substitute at Hilton High School in Hilton, NY, a rural suburb of Rochester, where I teach geometry. I will likely teach in similar schools in the future, so I have constructed my fictional course with my current students in mind. Across my three Regents geometry classes, I teach 67 students with a variety of identities: 33 males, 34 females, three students with IEPs, five students with 504 plans, 26 9th graders, 40 10th graders, and one 11th grader. All of my Regents geometry students are taught entirely in person and receive three 58 minute lessons per week plus one 35 minute lesson on Fridays. The student body is predominantly White, with only about 13% of students being from ethnic minority groups according to US News and World Report (“Hilton High School,” n.d.). Additionally, roughly 27% of the students come from economically disadvantaged backgrounds (“Hilton High School,” n.d.). According to the United States Census Bureau, over 95% of adults in Hilton age 25 and up have a high school diploma, but only 27% have a Bachelor’s degree or higher (“QuickFacts: Hilton village, New York,” n.d.). Data USA shares that the five most common types of occupations in Hilton are office and administrative support, sales/sales related, production, food preparation and serving, and transportation roles, which in total account for almost 54% of the jobs held by Hilton residents (“Hilton, NY,” n.d.). Politically, it is hard to tell where the suburb falls: According to BestNeighborhood.org, Hilton, NY leans slightly Democratic based on election results (“Hilton, NY Political Map – Democrat & Republican Areas in Hilton,” n.d.). However, conversations with veteran teachers at the school tell me otherwise; they share many stories of retaliation from parents regarding potentially left-leaning comments in classrooms, progressive curriculum, or actions of students who are exploring their

gender identity. Additionally, I have heard students make conservative and pro-Trump comments in the classroom and to teachers. I have kept all of this in mind while developing my algebra 1 mini-unit.

This mini-unit explores the world applications of exponential functions. At Hilton High School, exponential functions are typically taught right after Thanksgiving break. This is still fairly early in the school year, but students have had the time to familiarize themselves with content such as linear functions and systems of linear functions, explore how to graph equations, understand the meaning of variables, and determine how equations represent a related change between an independent and dependent variable, as well as the school culture and classroom expectations. Each student enters the classroom with a different perspective, but the reality in Hilton is that many students' perspectives are shaped by similar experiences: a lot of these students have grown up in similar cultures, in families of similar socioeconomic status, and attended the same middle and high schools, the only public options in the area. While some students care about their academic success, the motivation seems to stem from a desire to get to what is next; they have to do what they have to do in order to go to college, get the job they want, make a certain amount of money, or just get out of high school, whatever their goal may be. Math is just a cog in the wheel of required academics, and it gets significantly more challenging as the years go on. By the time students reach high school, they have usually aged out of asking their parents for help with their math homework; few people really remember what they learned in high school math classes, unless they studied math after high school or pursued a STEM related career, and even then it is unlikely they know the content in such a way that they could teach it. Based on the occupation statistics for the area, most of these students' parents probably do not work in fields that require a high-level understanding of mathematics, and so the

seemingly abstract math concepts taught in high school seem unnecessary for the real world. For students, high school math starts to feel more like a chore than a learning experience, and when the adults they look up to seem to have gotten by just fine without remembering any of their high school math, students struggle to see the point of it. My goal for this mini-unit is to show how math is interwoven into everyday life. We might not notice it or need to understand it on a deep, theoretical level to do our day-to-day activities, but we live and breathe math all the time, and foundational math skills are a tool to understanding our world. My students might not wish to be statisticians, engineers, actuaries, or analysts, but math will help them with the things that matter to them. This mini-unit is designed to develop confidence in students' math identities, connect relevant worldly topics to mathematics, and encourage the growth of critical thinking and analysis skills by being able to read our world through a math lens.

Unit Description and Purpose

Essential Question: What are exponential functions and how can they be used to depict our world?

This essential question targets not only the math standards related to exponential functions in algebra 1, but also the relevance of this mathematical concept in all our lives. In this mini-unit, students will learn how to write, graph, and interpret exponential functions. There will be an introductory, skill-building lesson intended to develop the mathematical skill required for understanding and using exponential functions, but thereafter, lessons will be centered around a critical issue each student has encountered or will encounter in their lives: the COVID-19 pandemic, climate change, and personal finances. The intention of framing these lessons around

different real-world topics is to make math interdisciplinary and peak students' interest in math through alternate topics; some students may already have interest in these topics, while others may be less informed in these topics but may find them interesting after exposure to them through these lessons. Throughout these lessons, we will also explore the stories that are being told about each of these critical issues through the presentation of the data in a particular model, critically assess why people might choose to present data a particular way, and theorize what other stories could be told if the same data was modeled differently. There will also be a lesson based on the children's book *One Grain of Rice* by Demi. Students will make predictions about the situation presented in the book, write an exponential function to model the situation after listening to the full story, and then reflect on the intuitiveness of the behavior of exponential functions. This lesson is designed to further examine the narratives that mathematical models tell via a literal story. Additionally, it introduces how our intuition can sometimes disagree with the reality of a situation and therefore we must be mindful of our original perceptions and critical of the presentation of data. The mini-unit culminates in a project where students analyze a non-math topic or social issue through an exponential model. They get to pick this topic entirely on their own based on their own interests, with the only requirement being that it can be modelled using an exponential function. Students are tasked with finding a model that relates to their topic of interest, explaining the model to their peers, sharing what they learned about the topic or issue, and drawing a conclusion about how the story of this topic or issue could look different if the data were modeled differently. In all, this unit is designed to show students that math is more than sitting at a desk completing computations, and that it actually plays a big role in a lot of the things we care about.

Literacies Framework (Five Pursuits)

The Pursuit of Identity

This unit supports the pursuit of identity by empowering students to identify themselves as mathematicians who can and deserve to understand math. In *Cultivating Genius*, Gholdy Muhammad calls on teachers to, “Ask students to tell you in their own words how they see their various identities. And when they tell you who they are, listen and trust them” (Muhammad, 2020, p. 71). Having students write their own math stories is a way of allowing them to share their identity with us specifically as it relates to their relationship with mathematics. They can write about their family, friends, school experiences, culture, religion, age, where they live, experiences they’ve had, or anything else of their choosing, with the only requirement being that they relate it to how they feel about math. This assignment tells students that the context of who they are matters in the math classroom, and that who they are, how they view themselves, and what their relationship is to math does not make them any more or less capable of doing math or deserving of the title of “mathematician.” This assignment resembles Muhammad’s suggested name stories activity and, like name stories, I believe the math stories activity will allow students “along with teachers, [to] begin to learn more about who they are” (Muhammad, 2020, p. 76).

Students will continue to explore and express their identities through the Graph of Me activity, which allows students to create a visual/mathematical representation of themselves through graphing. While this activity is not nearly all-encompassing of students’ identities and intentionally does not ask students to answer questions that require a ton of vulnerability (I did this so that students would be more likely to participate), it does allow them to see parts of themselves within a math model. Students also get an added layer of identity exploration by being given time to decorate the graph however they’d like. This shows students that math

activities can be expressive. When I hang the graphs up on the walls, students will be able to see the diversity within the classroom by the visual differences between the graphs. They will also hopefully gain a sense of belonging as they see themselves represented on the walls of their classroom.

The Pursuit of Skills

Muhammad defines skills as the “competence, ability, and expertise based on what educators deem to be important for student learning in each content area” (Muhammad, 2020, p. 85). This course is designed for algebra 1 in New York State, so these skills are best defined by the applicable NYS Next Generation Mathematics Learning Standards for algebra 1. These standards are primarily AI-A.CED.1 - “create equations and inequalities in one variable to represent a real-world context,” and AI-A.CED.2 - “create equations and linear inequalities in two variables to represent a real-world context” (New York State Education Department, 2017, p. 113). Occasionally, we may touch upon AI-A.CED.4 - “rewrite formulas to highlight a quantity of interest, using the same reasoning as in solving equations” (New York State Education Department, 2017, p. 113). In simpler terms, students will learn how to read, write, and interpret exponential functions including exponential growth/decay and compound interest, how to graph and interpret the graphs of these functions, and how to answer word problems related to this topic.

The Pursuit of Intellect

This mini-unit is not only designed to teach students about algebra 1, but also about relevant worldly topics that algebra can be used for. In this way, these lessons foster the pursuit of intellect. Muhammad defined intellect as “the understanding, enhancement, and exercising of mental powers and capacities that allow one to better understand and critique the world” (Muhammad, 2020, p. 104). In these lessons, students exercise their understanding of exponential functions to better understand real-world issues such as COVID-19 death rates, climate/environmental changes, and personal finance. Through this mini-unit, students will become more knowledgeable of these real-world topics and be able to connect and utilize their understanding of algebra to further investigate these topics.

The Pursuit of Criticality

A major part of this mini-unit is the exploration of what narratives are being pushed by certain mathematical models. Math is a powerful tool to help display and analyze data, but math models are even more powerful in that they can be used to display data in a *certain way*. That certain way of displaying data can alter the message the data is telling or cause people’s perceptions of the topic to lean a particular way. Starting with the Graph of Me activity, students will be asked to critically analyze mathematical models and the stories they may be telling. In this activity, students will determine if they think the graph they created is truly representative of them, or if there is more to their story than just that mathematical representation. Additionally, students will be tasked with comparing their graphs to others and exploring how different models tell different stories.

As the mini-unit progresses, the pursuit of criticality is more explicitly present and in direct line with Muhammad's definition of criticality: "the capacity to read, write, and think in ways of understanding power, privilege, social justice, and oppression, particularly for populations who have been historically marginalized in the world" (Muhammad, 2020, p. 120). Lessons 2, 4, 6, and 8 are designed around having students investigate exponential functions in relation to social justice issues. During lesson 2, students will read an excerpt from an article which discusses how math is political and how racism can often be found lying within data. Lesson 4 discusses the difference in death rates due to COVID-19 between African American Californians and White Californians, and then asks students to consider possible reasons why African American Californians had a higher death rate, which is intended to shed light on systemic racism. Lesson 6 discusses the pressing social justice issue of climate/environmental change, and also has students reflect on the exponential growth bias (the idea that we do not intuitively understand exponential growth) and the impact this bias can have on our perception of data. Finally, the summative assessment for this mini-unit asks students to pick a real world issue or topic of interest that can be represented using exponential functions, explain the model, analyze and critique the narrative being told by the particular model, and offer an alternative story that could be told with the data. This project encourages students to explore social justice issues, and regardless of if they choose a social justice related topic, they are asked to pursue criticality with their analysis of the narratives being pushed by the model. All students get to learn about the relevance of math in these topics and get to be critical of the narratives since these projects are going to be presented to the entire class during lesson 8.

The Pursuit of Joy

The pursuit of joy is, by far, my favorite literacy pursuit and is one I have tried to incorporate into every lesson. In *Unearthing Joy*, Muhammad states that joy is about fun and celebration, but in addition to that, it is “the embodiment of, learning of, and practice of love of self and humanity, and care for and help for humanity and the earth... Joy is also related to advancing happiness by elevating beauty in humanity, as well as embracing truthful narratives and representations of diverse people of the world (including our students)” (Muhammad, 2023, p. 69). Based on this definition, I think the pursuit of joy is present whenever people are enjoying the activity they are doing or the topic they are learning about, engaging in a loving practice that acknowledges people’s humanity, and are representing themselves and/or learning about others. By these standards, the pursuit of joy is present in all of my lessons; from writing math stories in lesson 1 to creating and decorating our own personal graphs in lesson 2 to regularly exploring worldly topics and social issues related to ourselves and others to finally sharing our own projects based on our interests in lesson 8, students can find something that sparks joy for them in each lesson.

Lesson Plans: Overview

Mini-Unit: Exponential Functions in Our World

NYS Next Generation Mathematics Learning Standards - Algebra I:

Creating Equations (A.CED)

- AI-A.CED.1 Create equations and inequalities in one variable to represent a real-world context.

- AI-A.CED.2 Create equations and linear inequalities in two variables to represent a real-world context.
- AI-A.CED.4 Rewrite formulas to highlight a quantity of interest, using the same reasoning as in solving equations.

Essential Questions:

Key Essential Question: What are exponential functions and how can they be used to depict our world?

Additional related questions:

- How are linear and exponential functions similar? How are they different?
- What do exponential functions look like on a graph?
- Where do we see exponential functions in our world?
- What story do exponential functions tell about a situation?

Lesson Sequence:

- Lesson 1 - We Are All Mathematicians!
- Lesson 2 - Accurate or Inaccurate?: Math as a Tool for Representation
- Lesson 3 - Introduction to Exponential Functions
- Lesson 4 - COVID-19: Understanding the Pandemic through a Mathematical Lens
- Lesson 5 - Our Perception of Exponential Models: How Fast Do They Really Grow?
- Lesson 6 - Our Perception of Exponential Models and How it Might Impact Our Interpretation of Climate Change
- Lesson 7 - Exponential Functions as a Tool in Personal Finance

- Lesson 8 - Exponential Functions in Our World Project Presentations

Monday	Tuesday	Wednesday	Thursday	Friday
				Lesson 1
Lesson 2	Lesson 3	Lesson 4	No Class	Lesson 5
Lesson 6	Lesson 7	Lesson 8		

Lesson 1 - We Are All Mathematicians!

Learning Goals and Desired Results:

- Students will be able to identify parts of their identities or personal histories that have shaped their relationship with math.
- Students will be able to identify similar and different experiences with and feelings about math among their classmates.
- Students will be able to identify themselves as mathematicians, regardless of their prior math experiences.
- *Pursuits: identity and joy*

Assessment:

To encourage vulnerability and genuine self-reflection, students will not be required to submit their math stories or discuss them in explicit detail during class. However, student participation during the four corners activity will be tracked and students will be asked to complete a brief reflection exit ticket about what they learned during the class.

Class Structure:

- *Total lesson - 35 minutes (Friday period, short block)*
- I will share my math story with the class (*3 minutes*)
 - My mom and my grandpa were both math majors, so math is “in my blood.”
 - I hated math in elementary school because I thought I was supposed to hate math since everyone else did.
 - In 5th grade, I realized I was pretty good at math and started to like it.
 - Math was kind of easy for me through high school.
 - In college, I wanted to study theater, but I studied math as well so I could possibly be a math teacher one day, especially if being an actress didn’t work out.
 - I ended up not liking studying theater, so math became plan A instead of plan B.
 - College math was very challenging.
 - Key experience: Being the only girl and only junior in an advanced math elective (the three other students were all seniors, boys, and engineers). I had to work so hard to try to understand the content, but now I am really proud that I made it through that course.
 - I’ve discovered that struggling is good and that math can be very fun and rewarding! Now I hope to teach my students that as well.
- Student math story - independent writing (*10-12 minutes*)
 - Students will write their own math stories, reflecting on their identities, histories, and prior experiences with math. The assignment is based on the [“What’s Your Math Story?” activity by Amber Fornaciari](#), and her suggested prompts are used on the assignment sheet.

- 4 Corners activity (*15 minutes*)
 - During the independent writing activity, I will place signs in each corner of the classroom, labeling each corner “Strongly Agree,” “Slightly Agree,” “Slightly Disagree,” or “Strongly Disagree.”
 - When the students finish their math stories, they will stand up at their desks and the game will begin.
 - I will say a series of statements which will prompt discussion about our math stories (listed below). The students should move to the corner of the room that best matches their response to the statement. Students will then discuss why they have chosen to stand in that given corner with their peers who chose the same answer. After roughly 2 minutes of discussion, students will elect a peer from their group to summarize their discussion: Were there any common reasons the students chose to stand in that corner? How did their reasons differ?
 - Repeat these steps for all of the statements or until the allotted time (about 15 minutes) has passed.
 - I hate math.
 - Math is easy for me.
 - Everyone I know thinks math is hard.
 - Everyone I know hates math.
 - My opinions about math have changed over time.
 - I enjoy the challenge of solving math problems.
 - I get frustrated when I can't solve math problems.
 - I want to keep studying math in the future.

- I think math has a lot of value in understanding our world.
- Exit Ticket/Homework (*5 minutes*)
 - Students will submit a brief reflection about their takeaways from the lesson.
 - If there is enough time after the 4 Corners activity, students will be given the opportunity to complete this reflection in class. Otherwise, this is to be done for homework and should take approximately 5 minutes.

Materials:

[My Math Story Activity](#)

[Exit Ticket/Homework Google Form](#)

Lesson 2 - Accurate or Inaccurate?: Math as a Tool for Representation

Learning Goals and Desired Results:

- Students will understand how math can convey a variety of information, and in some cases convey information incorrectly.
- Students will review their knowledge of coordinate planes and plotting points.
- Students will be able to draw conclusions based on visual representations on a graph.
- Students will understand the potential dangers of accepting mathematical models as fact rather than as tools for representation and storytelling.
- *Pursuits: identity, criticality, and joy*

Assessment:

- Students will demonstrate their knowledge of coordinate planes, plotting coordinate points, and reading graphs through their participation in the Graph of Me activity.
- Students will share their understanding of mathematical misinformation and math's use as a tool for conveying information through the exit ticket, completion of their homework, and their participation in lesson 3's class discussion.

Class Structure:

- *Total lesson - 58 minutes (standard period)*
- Students will read a short excerpt from the article "Mathematics in Context: The Pedagogy of Liberation" by Marian Dingle and Cathery Yeh. They will then engage in a class discussion about the reading prompted by some or all of the following questions:
(15 minutes)
 - Do you agree that math is political?
 - Why might people choose to convey mathematical information/use math tools in a particular way?
 - Do you believe students will learn better if the content is related to what they are interested in?
- I will introduce the Graph of Me activity by sharing my own graph with the class. Students will complete the activity independently *(15 minutes)*.
- As I hang the completed graphs around the classroom, students will watch the 15 minute video "When Bad Math Spreads Misinformation" by Math The World. They will then

complete an exit ticket answering the question, “Why is it important to analyze mathematical models and be critical of the stories they tell?” (20 minutes)

- The class will conclude with a discussion about what conclusion can be drawn based on looking at the graphs we created about our identities. Students will be asked to assess if they think their graph accurately represents who they are in their entirety, or if it is only telling a particular story.
- Homework: Students will select a graph, table, or other mathematical model related to a social issue they are interested in and share it using a Google Form.

Materials:

[Article Excerpt](#)

[Graph of Me Activity](#)

[“When Bad Math Spreads Misinformation” Video](#)

[Exit Ticket](#)

[Homework](#)

Lesson 3 - Introduction to Exponential Functions

Learning Goals and Desired Results:

- Students will be able to identify real-world situations that can be modeled using exponential functions.
- Students will be able to critically evaluate data to determine what message it is telling.
- Students will be able to communicate their analysis of data to their peers.

- Students will be able to define and use the following vocabulary:
 - nonlinear
 - exponential function
 - parent function
 - vertical stretch
 - vertical shrink
- Students will be able to identify parts of exponential functions and determine how they impact the outputs of the function.
- Students will be able to interpret the graphs of exponential functions.
- *Pursuits: skills, criticality, and joy*

Assessment:

To assess their understanding of the skills explicitly taught in the lesson (writing, evaluating, and graphing exponential functions), students will complete the Kuta Software Exponential Functions worksheet during class, which will be collected at the end of the period. Additionally, the summative assessment, the Exponential Functions in Our World project, will be introduced during this class.

Class Structure:

- *Total lesson - 58 minutes (standard period)*
- Students will spend the first 5 minutes of class sharing their selected graph, table, or other mathematical model related to a social issue with a partner. In their conversations, they will be asked to discuss the following prompt:

- What narrative is being shared through this mathematical model? How might this story look and sound different if the same data was presented using a different model?
- I will lead the class in a skill-based lesson that introduces exponential functions. This lesson is taught using the class notes packet I developed, which utilizes definitions, explanation, and questions from chapter 6 of Bright Ideas Math Algebra I: A Common Core Curriculum by Ron Larson and Laurie Boswell. This lesson will cover pages 1 through 6 of the notes packet. *(30 minutes)*
- Students will independently complete the Kuta Software Exponential Function worksheet, which will be collected and graded. *(10 minutes)*
- Introduce the Exponential Functions in Our World project *(3 minutes)*
 - Students will receive an assignment sheet with the project requirements, due date, and a grading rubric. I will walk students through the assignment sheet and take any questions they have.

Materials:

[Exponential Functions Notes Packet](#)

[Kuta Software Exponential Function Worksheet](#)

[Exponential Functions in Our World Project Assignment Sheet](#)

Lesson 4 - COVID-19: Understanding the Pandemic through a Mathematical Lens

Learning Goals and Desired Results:

- Students will understand the behavior of exponential functions on a graph.
- Students will be able to compare two exponential curves at a time and determine their similarities and differences.
- Students will understand how scaling a graph affects the look of the graph.
- Students will be able to critically assess how the look of a graph influences our perception of the data.
- *Pursuits: skills, intellect, criticality, and joy*

Assessment:

Students will be qualitatively evaluated on their participation in the class discussion. They will also begin working on their summative assessment, the Exponential Functions in Our World project.

Class Structure:

- *Total lesson - 58 minutes (standard period)*
- As a class, read aloud the excerpt from “Social Justice and Exponential Functions: Using Pandemic Data to Increase Student Understanding” by Jane Friedman (2022), which is highlighted in yellow on pages 302-304. *(5 minutes)*
- In groups of 3 or 4, have students explore the exponential functions provided in the excerpt using Desmos (this activity is somewhat of a mini reenactment of the activity

described in the article). Ask students to investigate the following questions and record their thoughts on paper (one worksheet per group):

- What happens when you make the range of the x-axis large (ex. $-5 < x < 100$) and the range of the y-axis is really small (ex. $-0.1 < y < 0.1$)? What story does this tell about the relationship between the COVID death rates of African American Californians versus White Californians?
- What happens when you make the range of the x-axis small (ex. $-5 < x < 40$) and the range of the y-axis large (ex. $-0.1 < y < 0.5$)? What story does this tell about the relationship between the COVID death rates of African American Californians versus White Californians?
- How do the two stories discovered in questions 1 and 2 compare?
- The curves intersect when $x=92$, after which the White Californians have a higher death rate. Why isn't this very meaningful? Explain why the smaller values of x are more significant in this story than the larger values.
- Why might the COVID death rate of African American Californians be higher than that of White Californians?
- After about 15 minutes of group work, the class will discuss the answers to these questions.
 - Ensure that question 4 is discussed and students realize that x represents age. If the curves do not intersect until $x=92$, this means that African American Californians have higher death rates than White Californians up until people turn 92 years old, when White Californians start to have higher death rates. 92 is already very old and many people are close to dying at that point, so comparing

the death rates of such elderly people is not very significant. The death rates at younger ages (smaller values of x), when people are generally less likely to die, say more about the racial disparities in COVID's impact.

- Students will utilize the remainder of class to work on their projects.

Materials:

[Friedman 2022 COVID Data Article](#)

[COVID Data Investigation](#)

Lesson 5 - Our Perception of Exponential Models: How Fast Do They Really Grow?

Learning Goals and Desired Results:

- Students will investigate the behavior of exponential functions.
- Students will understand the rate at which exponential functions grow and compare it to the rates of linear functions.
- Students will be able to construct an exponential function to match given data.
- *Pursuits: skills, intellect, and joy*

Assessment:

We will begin listening to the reading of *One Grain of Rice* by Demi. After the situation is introduced, students will write down a hypothesis on how many grains of rice they think will be given on day 30 if the number of grains is doubled every day. After finishing the story, they will work in pairs to develop an exponential formula to model the situation presented. We will then

go over the formula to ensure all students understand why the correct formula is $y = 2^{x-1}$.

Finally, students will reflect on if their hypothesis was correct and if this exponential growth was intuitive for them. This classwork will be collected.

Class Structure:

- *Total lesson - 35 minutes (Friday period, short block)*
- Watch the reading of *One Grain of Rice* by Demi.
- Pause the video at 3:56 and have students answer question 1 on their worksheets:
 - How many grains of rice do you think Rani will receive on day 30 if she receives one grain on day 1 and each day thereafter the number of grains is doubled?
- Continue watching the video.
- At the end of the video, have students work in pairs to develop an exponential function that models the situation in the book. *(10 minutes)*
- Go over the correct function, $y = 2^{x-1}$.
- Students will respond to question 3 and then submit their work:
 - Did you think Rani would receive this much rice on day 30? How accurate was your guess? Was the growth of the number of grains of rice intuitive, or were you surprised by this result?

Materials:

[One Grain of Rice Read Aloud](#)

[One Grain of Rice Mathematical Model Worksheet](#)

Lesson 6 - Our Perception of Exponential Models and How it Might Impact Our Interpretation of Climate Change

Learning Goals and Desired Results:

- Students will be able to define and use the following vocabulary:
 - exponential growth
 - exponential decay
 - growth/decay factor
 - rate of growth/decay
- Students will be able to solve word problems related to exponential growth/decay.
- Students will understand the importance of accurately perceiving the behavior of exponential functions.
- *Pursuits: skills, intellect, criticality, and joy*

Assessment:

Students will submit a written reflection about their reactions to the Exponential Growth Bias article excerpt. They will then be qualitatively evaluated on their participation in the class discussion. If time allows, students will also continue working on their summative assessment, the Exponential Functions in Our World project.

Class Structure:

- *Total lesson - 58 minutes (standard period)*
- I will guide students through a skill-based lesson on exponential growth and decay, which builds on the previous lesson about exponential functions. This lesson is also taught using

the class notes packet I developed, which utilizes definitions, explanation, and questions from chapter 6 of Bright Ideas Math Algebra I: A Common Core Curriculum by Ron Larson and Laurie Boswell. This lesson will cover pages 7 through 9 of the notes packet.
(15-20 minutes)

- After the lesson, students will independently read an excerpt from “Exponential Growth Bias and the Law: Why Do We Save Too Little, Borrow Too Much, and Fail to React on Time to Deadly Pandemics and Climate Change?” (Teichman & Zamir, 2022, pp. 1353-1354). They will then answer the following reflection question and submit their response:
 - How might misunderstanding or incorrectly predicting the outcomes of exponential functions lead to unjustified conclusions about worldly issues such as climate change?
- Students will utilize the remainder of class to work on their projects.

Materials:

[Exponential Functions Notes Packet](#)

[Exponential Growth Bias Reading and Reflection](#)

Lesson 7 - Exponential Functions as a Tool in Personal Finance

Learning Goals and Desired Results:

- Students will be able to define and use the following vocabulary:
 - principal

- annual interest rate
- compound interest
- Students will be able to model credit card debt and investments using exponential functions.
- Students will understand the utility of exponential functions in personal finance; they will know how these functions can positively (investing) and negatively (debt) impact their personal finances.
- *Pursuits: skills, intellect, and joy*

Assessment:

At the beginning of class, students will complete a discussion question worksheet based on the PBS video “The Math of Credit Cards.” At the conclusion of class, students will complete a brief exit ticket answering the question, “How can understanding exponential functions increase our financial literacy and help us plan for economically successful futures?” They will also continue working on their summative assessment, the Exponential Functions in Our World project, which they will present in the next class.

Class Structure:

- *Total lesson - 58 minutes (standard period)*
- Students will watch the PBS “The Math of Credit Cards” video and fill out the discussion question worksheet. *(10 minutes)*
- I will guide students through the Exponential Growth: Compound Interest section of the notes packet. This is a skill-based lesson teaching students about the exponential

functions that are relevant in personal finance. This lesson is also taught using the class notes packet I developed, which utilizes definitions, explanation, and questions from chapter 6 of Bright Ideas Math Algebra I: A Common Core Curriculum by Ron Larson and Laurie Boswell. This lesson will cover pages 10 and 11 of the notes packet. (25 minutes)

- Students will complete a sample problem about calculating compounding credit card interest.
 - This shows a negative perspective of compounding interest.
- Students will complete a sample question about compounding interest in a savings account.
 - This shows a positive perspective of compounding interest.
- Students will complete the personal finance exit ticket (5 minutes).
- Students will utilize the remainder of class to work on their projects.

Materials:

[PBS “The Math of Credit Cards” Video](#)

[PBS The Math of Credit Cards Discussion Questions](#)

[Exponential Functions Notes Packet](#)

[Exit Ticket](#)

Lesson 8 - Exponential Functions in Our World Project Presentations

Learning Goals and Desired Results:

- Students will be able to communicate the elements and behavior of exponential functions in a real-world context.
- Students will be able to critically assess the presentation of data.
- Students will learn about various real-world topics related to exponential functions.
- *Pursuits: skill, intellect, criticality, and joy*

Assessment:

Students will complete their summative assessment by presenting the findings of their project to the class in a roughly 3 minute presentation with a visual aid. They will be graded using the rubric on the assignment sheet.

Class Structure:

- *Total lesson - 58 minutes (standard period)*
- Students will each share a 3 minute presentation of their Exponential Functions in Our World project.

Materials:

[Exponential Functions in Our World Project Assignment Sheet](#)

Teacher Sample:

Lesson 2

Name: Teacher Sample
Date: _____

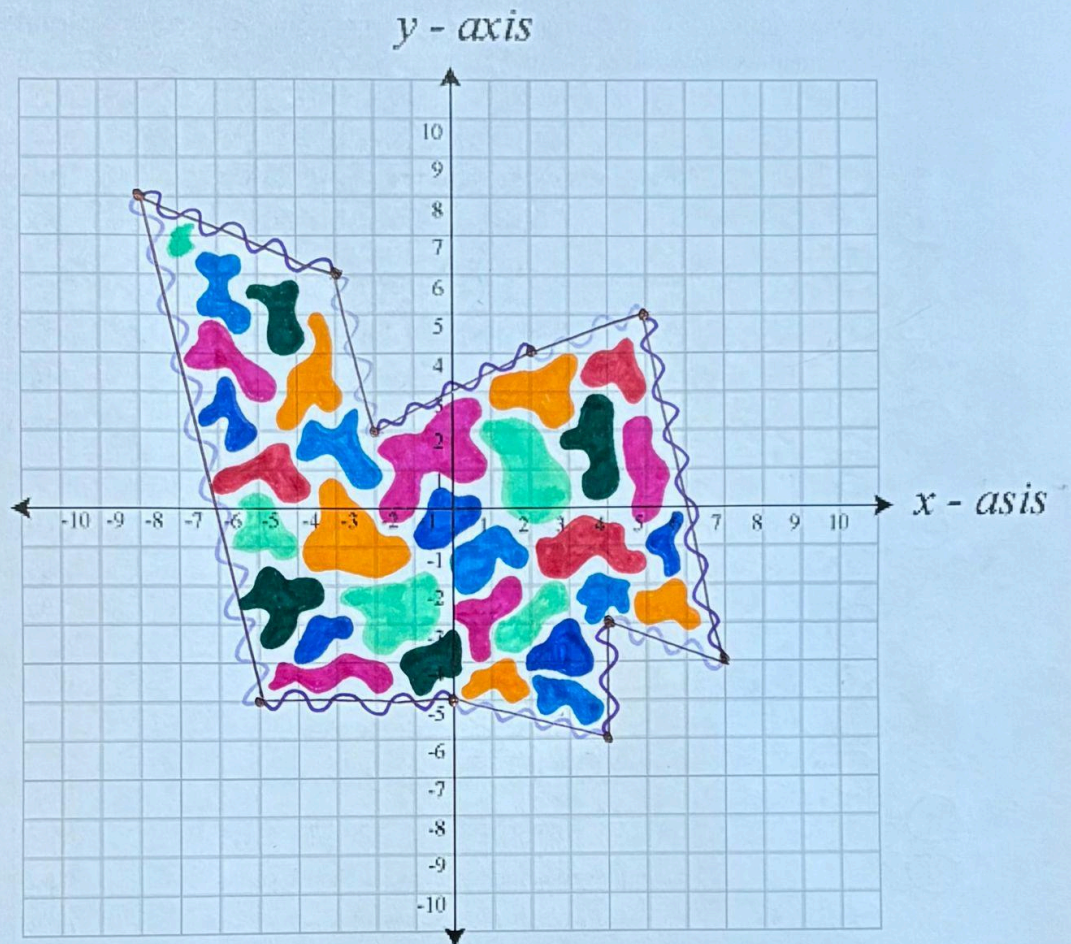
Algebra I
Graph of Me

Answer the true or false questions below. If you answer "true," plot the corresponding point on the coordinate plane on the next page. If you answer "false," do not plot the point. Once you have plotted all of the points for questions you answered "true," connect the dots on your graph, creating a unique figure that represents your answers. Then, decorate the figure however you'd like.

T	☒	I have lived in the same town my whole life.	(1,1)
☒	F	I am an only child.	(2,4)
T	☒	I have met all of my grandparents in my lifetime.	(3,-7)
☒	F	I have travelled outside of the United States.	(-2,2)
☒	F	I have a pet.	(-5,-5)
☒	F	I enjoy playing video games.	(7,-4)
T	☒	I speak more than one language.	(9,3)
☒	F	I can play a musical instrument.	(0,-5)
T	☒	I am ambidextrous.	(-3,0)
T	☒	I have broken a bone.	(0,8)
T	☒	I have an older brother.	(7,2)
T	☒	My favorite color is blue.	(3,-6)
T	☒	I can touch my tongue to my nose.	(-4,5)
☒	F	I have been on a road trip.	(-8,8)
☒	F	I have played a sport for more than 2 years.	(-3,6)
T	☒	My favorite subject is history.	(4,-8)
☒	F	Cooking is a big part of my family's culture and traditions.	(5,5)
☒	F	I am named after someone in my family.	(4,-6)
T	☒	I have a younger sister.	(0,2)
☒	F	I am an artist.	(4,-3)

Name: Teacher Sample
Date: _____

Algebra I
Graph of Me



Name: Teacher Sample
Date: _____

Algebra I
Exit Ticket

Accurate or Inaccurate?: Math as a Tool for Representation

Now that you have watched the video "When Bad Math Spreads Misinformation" by Math The World, answer the following question:

Why is it important to analyze mathematical models and be critical of the stories they tell?

We must be critical of math models because they can be inaccurate or misleading. We must check the sources of the data to ensure it is accurate and relevant to the goals of the model. A model can also be framed to intentionally make a story more or less dramatic. We can't allow ourselves to be convinced by the shock factor of certain numbers and models; we must first identify the source, accuracy, and framing of the information.



The Math of Credit Cards Discussion Questions

1. When you use a credit card, where does the money come from to pay for the goods you are getting?

Credit card companies

2. What is a line of credit?

The amount of money credit card companies are lending you.

3. What are the options for paying back a credit card loan?

all at once or a little at a time
(paid monthly)

4. What is a minimum payment?

The least amount you can pay in a month.
★ Doesn't pay your whole balance ★

5. What is interest?

% of your outstanding balance which will be added to your balance at the end of the month

6. How do credit card companies make their money?

Through interest (they get to keep the amount you pay in interest)

7. What are payday loans?

Loans with very high interest rates (200-400%)

↳ higher than minimum payments

8. Why do credit card companies not like people who are considered "dead beats"?

B/c they pay their balance each month, so the credit card companies don't make any \$ from interest.

9. What are the main lessons or advice given in the video?

1.) Avoid quick cash / payday loans

2.) Be a "dead beat" → pay in full + on time!

Exponential Growth: Compound Interest

Exponential growth functions are used in real-life situations involving *compound interest*. Although interest earned is expressed as an *annual* rate, the interest is usually compounded more frequently than once per year. So, the formula $y = a(1 + r)^t$ must be modified for compound interest problems.

Core Concept

Compound Interest

Compound interest is the interest earned on the principal and on previously earned interest. The balance y of an account earning compound interest is

total after interest $y = P \left(1 + \frac{r}{n} \right)^{nt}$

P = principal (initial amount)
 r = annual interest rate (in decimal form)
 t = time (in years)
 n = number of times interest is compounded per year

Examples: Credit Card Debt

- 1.) Charlie has a Chase Freedom Flex credit card, a very popular card amongst young adults. This card has a variable annual percentage rate (APR) of 18.24% - 27.74%. Right now, Charlie's APR is 20%, and his interest is compounded monthly. If Charlie has a starting balance of \$2,000 and does not make any payments, how much will he owe in 3 years?

$r = 0.2$ (20% in decimal form)
 $n = 12$ b/c there are 12 months in a year

$y = 2,000 \left(1 + \frac{0.2}{12} \right)^3$
 $y = 2,000 (1.0167)^3$
 $y = 2,000 \times 1.0508$
 $y = 2,101.6759$

\$2,101.68

- 2.) Shayla also has a Chase Freedom Flex credit card and her interest is compounded monthly, but her APR is 25%. If Shayla has a starting balance of \$1,800 and does not make any payments, how much will she owe in 3 years?

$r = 0.25$ $n = 12$

$y = 1,800 \left(1 + \frac{0.25}{12} \right)^3$
 $y = 1,800 (1.0208)^3$
 $y = 1,800 \times 1.0638$
 $y = 1,914.86002$

\$1,914.86

Example: Investing

- 1.) Emerson just opened up a Capital One 360 Performance Savings account, which has an annual percentage yield (APY) of 3.10% and is compounded quarterly. If Emerson opened the account with \$10,000 and doesn't add any money to it, how much will he have in 20 years?

$P = 10,000$ $r = 0.031$ $n = 4$
 $t = 20$ $\uparrow$ quarterly

$$Y = 10,000 \left(1 + \frac{0.031}{4}\right)^{20}$$

$$Y = 10,000 (1.00775)^{20}$$

$$Y = 10,000 \times 1.16696$$

$$Y = 11,669.6045$$

\$11,669.60

- 2.) Suppose Emerson only opened the account with \$5,000, but after 5 years he deposited another \$5,000 into the account. In this case, how much money will be in the account 20 years from the date he opened it?

First 5 years

$P = 5,000$ $t = 5$ $r = 0.031$ $n = 4$

$$Y = 5,000 \left(1 + \frac{0.031}{4}\right)^5$$

$$Y = 5,000 (1.00775)^5$$

$$Y = 5,000 \times 1.03936$$

$$Y = 5,196.7765$$

Years 6-20 (15 years)

$t = 15$, $r = 0.031$, $n = 4$

$P = \text{amount in account} + \text{deposit}$
 $P = 5,196.7765 + 5,000$
 $P = 10,196.7765$

$$Y = 10,196.7765 \left(1 + \frac{0.031}{4}\right)^{15}$$

$$Y = 10,196.7765 (1.00775)^{15}$$

$$Y = 10,196.7765 \times 1.1228$$

$$Y = 11,448.6691$$

Vocab Break!

principal - the initial amount of money in the account

annual interest rate - percent of interest for the whole year

→ usually compounded more than once per year! ← n

Compound interest - interest earned on the principal and on interest that was previously earned

Name: Teacher Sample
Date: _____

Algebra I
Personal Finance Exit Ticket

How can understanding exponential functions increase our financial literacy and help us plan for economically successful futures?

Understanding exponential functions helps us see how interest will accrue on debt or investment. Accruing interest on debt is bad b/c it makes us owe more \$, but accruing interest on investments is good b/c we make \$!

Personal Reflection

Planning this mini-unit was very inspiring and informative, but not free of challenges. At the beginning, I struggled with connecting a concept I had previously taught to the real world, and I couldn't think of ways to genuinely engage students in the pursuits of intellect, criticality, and even identity without doing this. Connecting math to something that students felt pertained to their lives seemed essential if I wanted to whole-heartedly implement Muhammad's culturally and historically responsive literacy framework, but having only taught Regents geometry and the beginning units of algebra 1 so far, I felt limited by my content options and needed to branch out into math topics I haven't explored yet as a teacher. I was immediately drawn to a topic related to mathematical models and graphs since they are so integral in communicating information both in

and outside of math related fields. However, I didn't originally know what topic to pick that would touch on New York State standards for the grades I will realistically teach one day. Most of the graphing and modeling topics are covered in elementary education, so I couldn't simply design a lesson covering mathematical models. Instead, I had to find a secondary level math standard that I could relate to math models.

I eventually realized that exponential functions are very useful for modeling real-world situations, which is why I picked this algebra topic. This certainly added a challenge to this project as I had to reencounter and reteach myself exponential functions. Additionally, since I haven't studied this in years and haven't taught it yet, I needed to find or create all my own materials. Finding an anchor text stumped me since I didn't have many math resources of my own, digital textbooks are limited, libraries didn't seem to carry any textbooks I was interested in, and purchasing textbooks would be expensive. While I didn't need to use a textbook as my anchor text, I did need a source of math information to base the skill parts of my lesson on, so this is what I was focused on finding as my anchor text and why I leaned towards textbooks. I think I overprioritized this when starting the project, which dramatically slowed my progress. Once I let go of the idea of finding a perfect anchor text and just let myself start planning the lesson based on what I actually wanted to achieve in the mini-unit, things started to come together. This process taught me that if one approach to a project isn't working, just pivot. There is no point continuously trying to fit a square peg in a round hole; it won't fit. Instead, try getting started somewhere else and come back to the point of struggle later with fresh eyes. Luckily, when I did this, I was able to find a free online textbook to pull definitions and sample problems from, which worked perfectly. I also like that this is the source I ultimately ended up using since it is available online for free, which makes it very accessible to students.

Muhammad's framework can be intimidating at first. The five pursuits do not always seem inherently integrated into academic content, especially in STEM subjects. However, creating this mini-unit made it clear that these pursuits are much more fundamental to quality education than you may think, and they are rather easy to incorporate into lessons. Finding ways to connect your topic to worldly topics and discussion of power and social justice may add an extra step to the lesson planning process, but the result is a lesson that is much more inclined to interest students and make them feel seen in the classroom, which is well worth the effort it takes to add this to lessons. Now that I've created an entire mini-unit utilizing Muhammad's framework, I think it will be hard for me to not think about these pursuits when I am creating lessons or notice their presence (or lack there of) in unit design.

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